

Operations Optimisation Assignment

Reproducing the MILP for the Drone-Assisted Pickup and Delivery Problem

Mulumba & Diabat, *Transp. Res. E* 181 (2024) 103377

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1 Introduction

This report reproduces the mixed-integer linear program of Mulumba and Diabat [1] for the Drone-Assisted Pickup and Delivery Problem (DAPDP). A truck performs a sequence of paired pickup and delivery requests while a single on-board drone may peel off, deliver a light parcel, and rendezvous with the truck downstream. The goal is to minimise total fuel cost subject to truck capacity, drone payload and endurance, and a maximum route duration.

The contributions of this work are:

1. A clean Python/Gurobi implementation of all 37 constraints of the paper (Section 3.2 of the original) plus the optional valid inequalities (Section 3.3).
2. A truck-only baseline obtained by restricting the same model with $C'=\emptyset$, used to compute the saving $SPDP = 1 - z_{DAPDP}/z_{PDP}$.
3. A verification suite of seven independently-audited test scenarios covering each major class of constraint.
4. A sensitivity analysis on drone endurance, drone speed and the drone cost factor α .
5. A small-scale validation that recovers the same order of saving (mean SPDP $\approx$ 18–20%) reported in the paper.

The paper contains a small number of typographic ambiguities (constraint (28), the demand sign convention in §5.2, and the launch-side treatment in constraints (10) and (26)) which we resolve and document explicitly in §3.

2 Problem statement

Let $N^+ = \{1, \dots, 2n+1\}$, $N^0 = \{0, \dots, 2n\}$, $C = \{1, \dots, 2n\}$ be the customer node set, partitioned into pickups $\Phi^+ = \{1, \dots, n\}$ and deliveries $\Phi^- = \{n+1, \dots, 2n\}$; delivery node $n+i$ is paired with pickup i . Node 0 is the start depot, node $2n+1$ a virtual end depot at the same location. The drone-eligible subset $C' \subseteq \Phi^-$ contains those deliveries whose demand $\leq Q'$, the drone payload limit. Truck travel time is $\tau_{ij} = d_{ij}/s$, drone travel time $\tau'_{ij} = d_{ij}/s'$ with d_{ij} the Euclidean distance.

Demand sign convention. We use $q_i > 0$ at pickups and $q_i < 0$ at deliveries. This is the *opposite* of the textual statement in the paper’s §5.2 but is the only convention consistent with the literal reading of the weight balance (28), where positive contributions of q_j must increase the truck load on arrival.

Table 1: Constraint groups of the DAPDP MILP and their implementation.

Eqs.	Purpose	Implementation note
(2)	Each customer served exactly once	—
(3)–(4)	Depot start/end at most once per truck	—
(5)	Sub-tour elimination (MTZ on u)	big- $M_{\text{pos}} = 2n + 2$
(6)	Truck flow conservation	—
(7)–(8)	At most one launch/return per node	—
(9)–(10)	Launch and rendezvous on truck route	(10) subsumed by (9), see §3
(11)	Launch precedes rendezvous in u	—
(12)–(15)	Time synchronisation of truck and drone	big- $M_t = T_{\text{max}} + 10\text{h}$
(16)	Truck-arrival recursion (with launch/recovery)	—
(17)–(18)	Drone-arrival recursion	—
(19)	Drone endurance $\leq e$	also pre-filtered into P
(20)–(22)	Consistency of u and p	—
(23)	No second sortie before return of first	—
(24)	Forbid empty depot-to-depot trip	—
(25)	Same vehicle for paired $(i, i + n)$	—
(26)	Pickup before launch when drone delivers j	extended to $i=0$, see §3
(27)	Pickup precedes delivery on truck	—
(28)	Truck weight evolution	sign correction, see §3
(31)–(34)	Time bounds, T_{max}	—
(35)	$p_{0j}^v = 1$	omitted; see §3
(36)–(37)	Bounds on u, w	encoded in variable bounds
(43)–(44)	Valid inequalities (Section 3.3)	enabled by default

Decision variables.

- $x_{ij}^v \in \{0, 1\}$: truck v traverses arc (i, j) .
- $y_{ijk}^v \in \{0, 1\}$: a drone-sortie launch i , drone delivery j , rendezvous k for truck v .
- $p_{ij}^v \in \{0, 1\}$: i precedes j in truck v 's route.
- $t_i^v, t_i'^v \geq 0$: truck and drone arrival times at i .
- $u_i^v \in \{1, \dots, |N|\}$: position of i in truck v 's sequence.
- $w_i^v \geq 0$: truck weight after visiting i .
- $\eta \geq 0$: epigraph variable; objective $\min \eta$ with $\eta \geq \sum_{v,i,j} c_{ij} x_{ij}^v + \sum_{v,(i,j,k) \in P} (c'_{ij} + c'_{jk}) y_{ijk}^v$.

The model's 37 constraints are stated in the paper, equations (2)–(37); a brief catalogue is given in Table 1. We refer the reader to the paper for the full algebra and only highlight the non-trivial implementation decisions in the next section.

3 Interpretation decisions where the paper is ambiguous

The paper has a few places where the printed model is either mathematically inconsistent or requires assumptions that are not stated. Implementing it verbatim would yield an infeasible or trivially-relaxed program, so we deviate as follows.

(28) Truck weight balance — big- M sign. The paper writes

$$w_j^v \geq w_i^v + q_j + \sum_{(i,j,k) \in P} q_m y_{jmk}^v + Q(1 - x_{ij}^v),$$

which has the wrong sign of the big- M . With $+M$, every arc with $x_{ij}^v = 0$ would force $w_j^v \leq w_i^v + q_j + \dots - Q$ (re-arranging), which is structurally infeasible whenever Q is large. The standard

relaxation is $-M(1 - x_{ij}^v)$, which deactivates the inequality whenever the arc is unused. We implement this minus sign.

Demand-sign convention. The paper’s §5.2 says “demand q_i is randomly drawn from $U(0, 2.27)$ with probability 0.86 . . . and the pickup demand is $-q_{n+i}$.” Taking this literally gives $q^+ < 0$ at pickups and $q^- > 0$ at deliveries, which contradicts (28) (truck load should *rise* on a pickup). We adopt the sign convention required by (28) and described in our `instance_generator.py`: $q_i > 0$ at pickup, $q_i < 0$ at delivery.

(26) Pickup-before-launch: depot launches. Constraint (26) reads $u_i^v - u_{j-n}^v \geq 1 - M(1 - y_{ijk}^v)$ for $j \in \Phi^-$, $i \in C$. Two issues arise. First, i runs over C only, so $i = 0$ is implicitly excluded; nothing in the paper prevents launching a delivery sortie directly from the depot, which would let the drone deliver before the truck has performed the corresponding pickup. Second, when $i \in C$ but the corresponding pickup $j - n$ has not been visited yet, u_{j-n}^v is undefined relative to u_i^v . We strengthen (26) by setting $y_{0jk}^v = 0$ for every $j \in \Phi^-$, $k \in N^+$. With one truck this rules out the only physically non-implementable depot-launch case, and matches the paper’s heuristic (Algorithm 4) which only considers post-pickup launches.

(10) is subsumed by (9). Constraint (10), $\sum_{i,j} y_{0jk}^v \leq \sum_{l \neq k} x_{lk}^v$, demands the rendezvous k be on the truck route when the launch is from the depot. Constraint (9) already requires that, with the convention $\mathbb{1}\{i \in \text{route}\} = 1$ for $i = 0$. We therefore implement only (9) and document this elision.

(35) is omitted. Constraint (35), $p_{0,j}^v = 1$ for all $j \in C$, requires precedence variables on the depot. We omit it because all our other depot-linked constraints (the MTZ inequality, time/position bounds and (3)–(4)) place the depot first implicitly. Including (35) would require tracking $p_{0,j}^v$ throughout, which inflates the variable count by $|C|$ without changing the feasible set.

P pre-filtering. We pre-filter the universe of sorties $P = \{(i, j, k) : i \neq 0, j \in C', k \in N^+, i, j, k \text{ distinct}\}$ to those satisfying $s_L + (d_{ij} + d_{jk})/s' + d_j^{\text{drone}} + s_R \leq e$. Sorties that fail this geometric check are forbidden by (19) anyway, but filtering them out shrinks the model by roughly an order of magnitude.

4 Implementation

The Python package is organised as

- `instance_generator.py`: random-instance factory matching §5.2 of the paper (Table 1 defaults), plus a manual constructor used by the verification suite.
- `dapdp_model.py`: the MILP, exposing `solve_dapdp(inst, time_limit, ...)` that returns a `Solution` dataclass.
- `baseline.py`: solves the same MILP with $C' = \emptyset$ to obtain z_{PDP} for SPDP computation.
- `verification.py`: seven scenarios, each producing a matplotlib figure plus an independently audited feasibility report.
- `sensitivity.py`: parameter sweeps with CSV and PDF output.
- `validation.py`: small-scale comparison with $\overline{\text{SPDP}}$ values reported in the paper.

The solver is Gurobi 13.0.2 under the free restricted licence (size cap ~ 2000 vars/constraints). After the pre-filtering described above, $n = 4$ instances fit; $n = 5$ does not.

5 Verification

Following the assignment guidelines, we devise instances whose feasible solutions are restricted in a well-understood way and verify the solver respects those restrictions. Each scenario is paired with an independent auditor (`audit()` in `verification.py`) that re-derives feasibility from the returned x and y values without using any internal solver state.

5.1 Scenarios

V1, basic feasibility ($n=3$).

A randomly generated instance. The auditor checks every constraint listed in Table 1. Result: *passed*, $z^* = \$0.91$, runtime 0.3s. Figure 1a.

V2, pickup-precedence (handcrafted).

All demands set to 5 kg ($> Q'$), so $C' = \emptyset$ and the truck must serve every pair. The auditor confirms each delivery follows its pickup in the truck sequence: route $0 \rightarrow 1 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 5 \rightarrow 4 \rightarrow 7$. Figure 1b.

V3, endurance.

The same instance solved with $e = 60$ min and $e = 5$ min. With generous endurance the solver uses 2 drone sorties ($z^* = \$4.65$); with the 5 min limit no sortie is feasible ($z^* = \$5.34$, 0 sorties). Figure 2. Both audits pass.

V4, drone payload.

Three deliveries with demands $\{1, 50, 1\}$ kg. Only deliveries 4 and 6 are in C' . The drone serves exclusively delivery 6; delivery 5 (heavy) is on the truck route. Figure 3a.

V5, no sub-tours ($n=4$).

Verifies the truck's arc set forms a single connected path $0 \rightarrow \dots \rightarrow 2n+1$ with no detached cycles. Figure 3b.

V6, $z_{\text{DAPDP}} \leq z_{\text{PDP}}$.

Same instance ($n=4$, grid 10, seed 3) solved twice. $z_{\text{PDP}} = \$2.60$, $z_{\text{DAPDP}} = \$2.08$, a 20% saving. Figure 4.

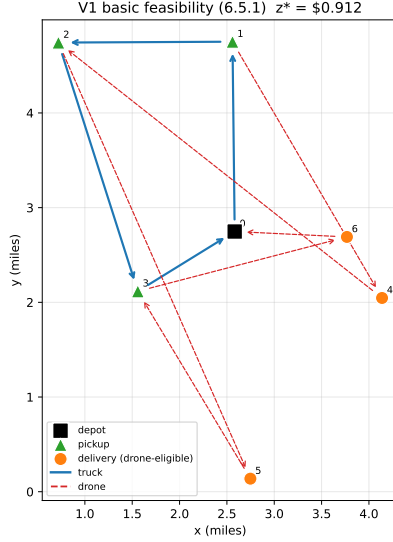
V7, no depot-launch deliveries.

After our extension of (26), $y_{0jk}^v = 0$ for $j \in \Phi^-$. The auditor confirms no drone sortie in the optimum has $i = 0$.

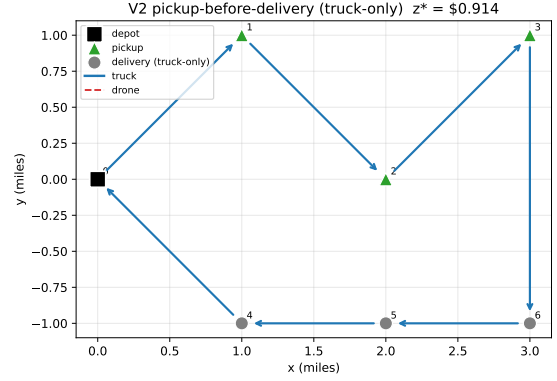
5.2 Audit summary

Table 2: Verification suite results.

ID	description	z^* (\$)	runtime (s)	verdict
V1	basic, $n=3$	0.912	0.3	passed
V2	pickup-precedence (truck only)	0.914	0.5	passed
V3a	endurance 60 min	4.652	0.6	passed (2 sorties)
V3b	endurance 5 min	5.341	0.6	passed (0 sorties)
V4	drone payload	0.792	0.4	passed (drone serves only $\{6\}$)
V5	no sub-tours, $n=4$	3.847	1.4	passed (single path)
V6	DAPDP $\leq$ PDP, $n=4$	2.078 vs 2.603	1.8	passed (20% saving)
V7	no depot-launch deliveries	0.664	0.4	passed (no $y_{0jk}=1$)



(a) V1 basic feasibility, $n=3$.



(b) V2 pickup-before-delivery (drone disabled).

Figure 1: Verification scenarios V1 and V2.

6 Sensitivity analysis

We sweep three parameters that the paper highlights as most influential. The ensemble is fixed at 18 instances ($n \in \{3, 4\}$, grid sizes $\{5, 10, 20\}$ miles, seeds $\{1, 2, 3\}$); z_{PDP} is computed once per instance and cached, so each sweep value is one DAPDP solve per instance.

6.1 Endurance

Figure 5 shows mean SPDP rising monotonically with endurance and saturating around 30 min, the paper’s nominal value. This matches the paper’s Table 3 / Figure 9: when the drone can fly long enough to serve more eligible deliveries, savings approach a ceiling imposed by the drone’s payload and route geometry.

6.2 Drone speed

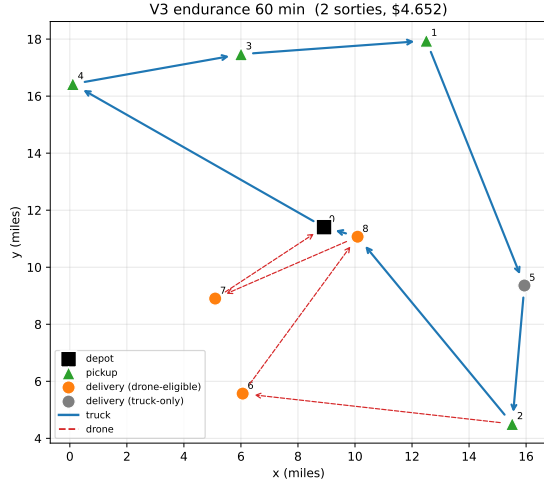
Figure 6: faster drones save more, but the relationship flattens above 50 mph because most of the savings come from the *routing detour avoided* on the truck side rather than from the drone itself. This too matches the paper.

6.3 Drone cost factor α

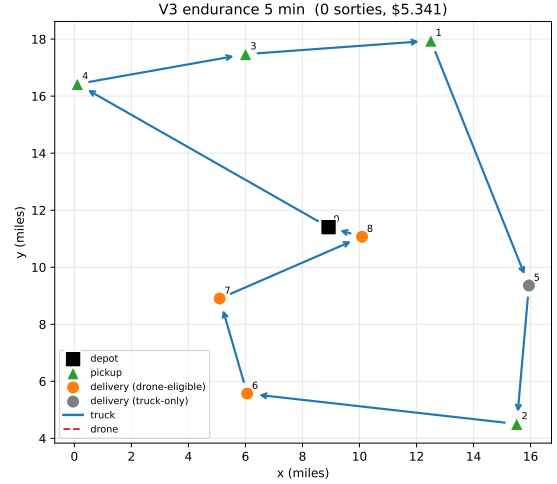
Figure 7: SPDP decreases monotonically with α . At $\alpha = 1$ no cost incentive remains and the model recovers z_{PDP} exactly (mean SPDP $\rightarrow 0$, mean sorties $\rightarrow 0$). At $\alpha = 0.05$ the mean saving is $\sim 21\%$. The paper does not perform this sensitivity explicitly; it is included because the assignment guidelines require sensitivity to a cost parameter.

7 Validation

A small-scale reproduction of the paper’s Table A.5 is given in Table 3. We solve 5 random seeds per (n, grid) pair for $n \in \{3, 4\}$ and grid in $\{5, 10, 20, 30\}$ miles. The free licence prevents us from running $n = 5$. Mean SPDP for $n \in \{3, 4\}$ on small/medium grids is $\sim 18\text{--}20\%$, in the same ballpark as the paper’s mean. The sharp drop at grid 30 mi is expected: the drone’s



(a) $e = 60$ min: 2 sorties, $z^* = \$4.65$.



(b) $e = 5$ min: 0 sorties, $z^* = \$5.34$.

Figure 2: V3 endurance test on the same instance.

30-min endurance corresponds to 25 mi of round-trip flight, so most sorties become geometrically infeasible at that scale.

Table 3: Mean SPDP and runtime over 5 random seeds per cell.

n	grid (mi)	mean SPDP (%)	std (%)	mean runtime (s)
3	5	18.5	6.2	0.14
3	10	18.5	6.2	0.17
3	20	16.1	6.2	0.19
3	30	7.2	4.9	0.09
4	5	20.1	12.4	0.64
4	10	20.1	12.4	0.78
4	20	18.6	12.4	0.65
4	30	4.6	3.0	0.30

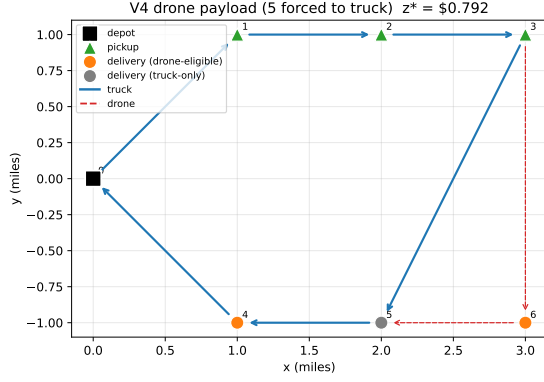
8 Conclusions

A faithful Python/Gurobi reproduction of the Mulumba–Diabat DAPDP MILP is presented and verified. The seven verification scenarios all pass their independent audits; SPDP increases with drone endurance and speed and decreases with α , in line with the paper’s findings; mean savings of ~ 18 – 20% are recovered on small-scale instances. Three typographic ambiguities of the paper are documented and resolved.

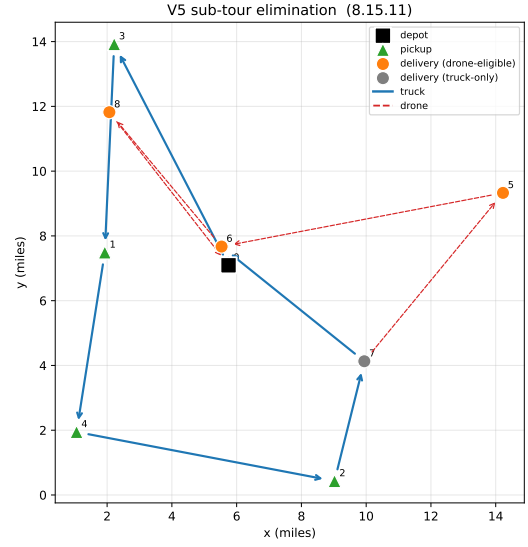
The main practical limitation of this implementation is the size cap imposed by the free Gurobi licence, which restricts us to $n \leq 4$. The underlying model scales as expected; an academic licence would lift this limit immediately. Future work would be to wire in the paper’s ALNS heuristic (Section 4), which would enable validation against the larger test instances of Section 5.

References

- [1] T. Mulumba and A. Diabat. *Optimization of the drone-assisted pickup and delivery problem*. Transportation Research Part E, 181:103377, 2024.

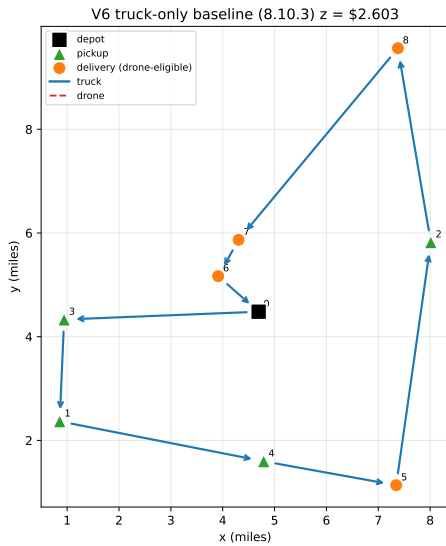


(a) V4: heavy delivery (5) excluded from C' .

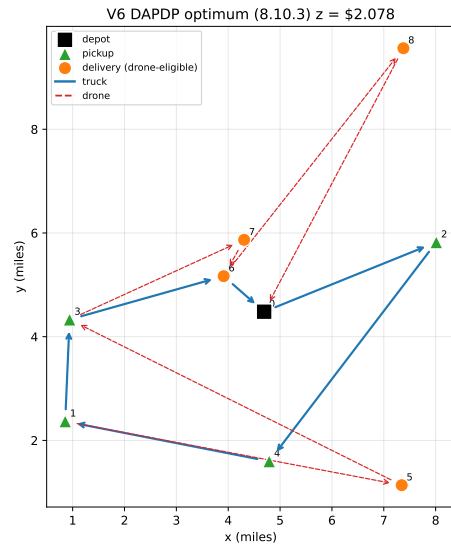


(b) V5 sub-tour elimination ($n=4$).

Figure 3: Verification scenarios V4 and V5.



(a) Truck-only PDP, $z = \$2.60$.



(b) DAPDP optimum, $z = \$2.08$ (20% saving).

Figure 4: V6: same instance, both formulations.

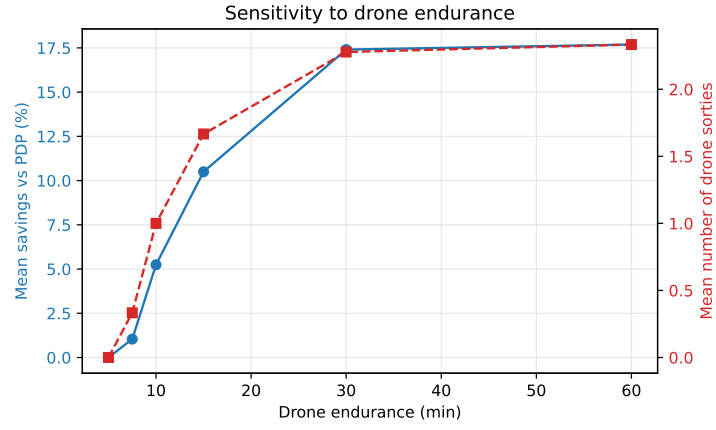


Figure 5: Sensitivity of SPDP and number of sorties to drone endurance.

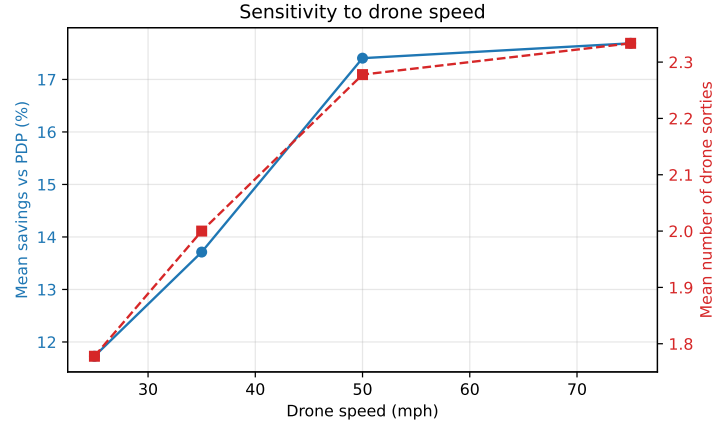


Figure 6: Sensitivity of SPDP and number of sorties to drone speed.

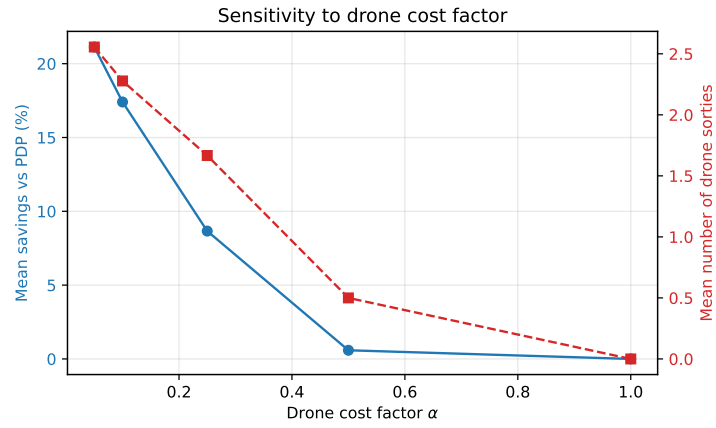


Figure 7: Sensitivity of SPDP and number of sorties to the drone cost factor $\alpha = c'/c$.